Sparks of Artificial General Intelligence: Early experiments with GPT-4

Sébastien Bubeck, Varun Chandrasekaran, Ronen Eldan, Johannes Gehrke, Eric Horvitz, Ece Kamar, Peter Lee, Yin Tat Lee, Yuanzhi Li, Scott Lundberg, Harsha Nori, Hamid Palangi, Marco Tulio Ribeiro, Yi Zhang
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Artificial intelligence (AI) researchers have been developing and refining large language models (LLMs) that exhibit remarkable capabilities across a variety of domains and tasks, challenging our understanding of learning and cognition. The latest model developed by OpenAI, GPT-4, was trained using an unprecedented scale of compute and data. In this paper, we report on our investigation of an early version of GPT-4, when it was still in active development by OpenAI. We contend that (this early version of) GPT-4 is part of a new cohort of LLMs (along with ChatGPT and Google's PaLM for example) that exhibit more general intelligence than previous AI models. We discuss the rising capabilities and implications of these models. We demonstrate that, beyond its mastery of language, GPT-4 can solve novel and difficult tasks that span mathematics, coding, vision, medicine, law, psychology and more, without needing any special prompting. Moreover, in all of these tasks, GPT-4's performance is strikingly close to human-level performance, and often vastly surpasses prior models such as ChatGPT. Given the breadth and depth of GPT-4's capabilities, we believe that it could reasonably be viewed as an early (yet still incomplete) version of an artificial general intelligence (AGI) system. In our exploration of GPT-4, we put special emphasis on discovering its limitations, and we discuss the challenges ahead for advancing towards deeper and more comprehensive versions of AGI, including the possible need for pursuing a new paradigm that moves beyond next-word prediction. We conclude with reflections on societal influences of the recent technological leap and future research directions.

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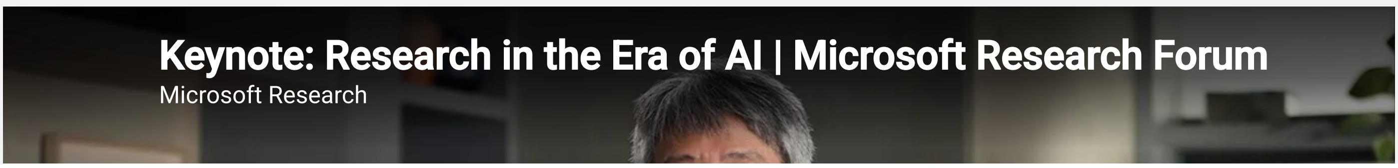

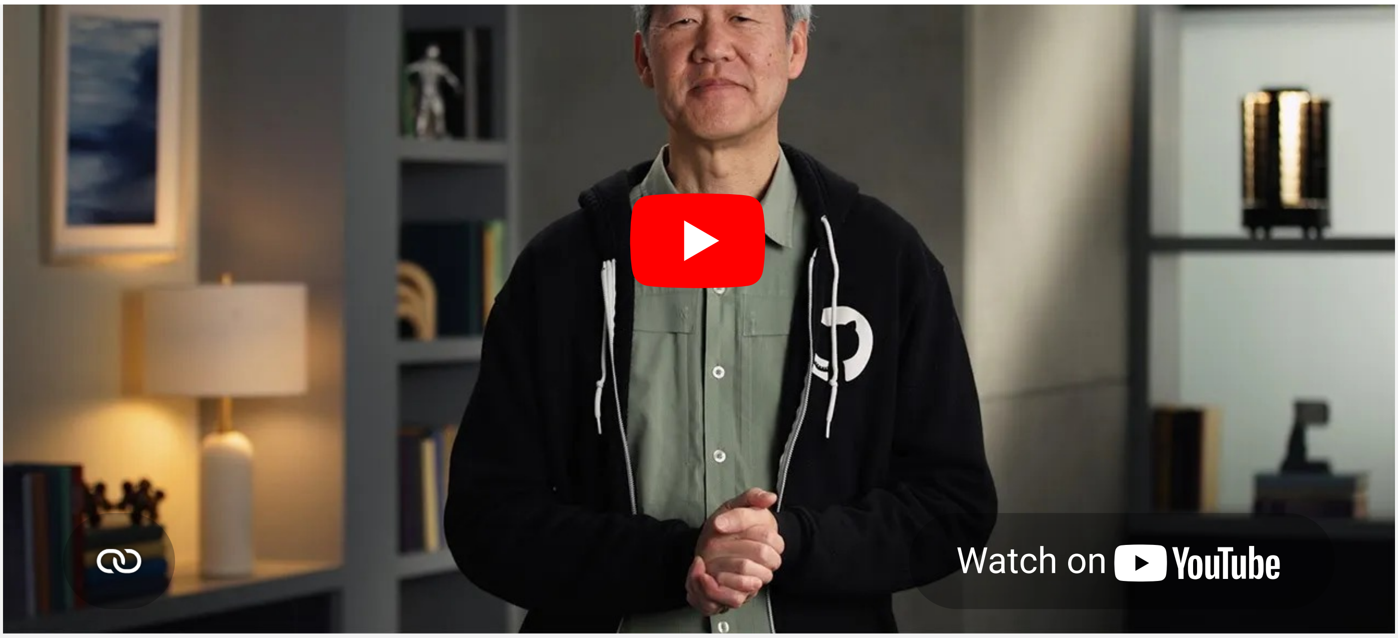

Keynote: Research in the Era of AI

*Presented by [Peter Lee] at **Microsoft Research Forum, Season 1, Episode 1***

Peter Lee, Corporate Vice President, Microsoft Research and Incubations, discusses how recent developments in AI have transformed the way Microsoft approaches research.

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Keynote: Research in the Era of AI

PETER LEE: Hi. I'm really pleased and excited to be here for this first Microsoft Research Forum, a series that we have here out of Microsoft Research to carry out some important conversations with the research and scientific community.

This past year has been quite a memorable one. Just some incredible advances, particularly in AI, and I'll spend a little bit of time talking about AI here to get us started. But before doing that, I thought I would try to at least share how I see what is happening in the broader context of scientific disruption. And to do that, I want to go all the way back to the 1700s and the emerging science of biology, the science of living things. Actually, in the 1700s, it was well understood by the end of that century that all living things were made up of cells—everything from trees and plants to bugs, animals, and human beings. But a fundamental scientific mystery that lingered for decades was, where do cells come from?

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